

TFG - 4. Anàlisi financera

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```
library(dplyr)
library(lubridate)
library(ggplot2)
library(zoo)
library(tidyr)
#install.packages("tidyquant")
library(tidyquant)
library(corrplot)
library(scales)
```

4.1. Anàlisi descriptiva i financera

Lectura de la base de dades i agrupació per mesos

```
dades_netes <- readRDS("C:/Users/joelm/Downloads/TFG/dades_netes.rds")
summary(dades_netes)
```

```
##      ID_Card      Date      Price      Vintage
## Length:20559    Min.   :2016-09-12    Min.   : 10.0    Moderna:10887
## Class :character 1st Qu.:2021-01-02    1st Qu.: 152.5    Vintage: 9672
## Mode  :character Median :2022-09-13    Median : 375.0
##                      Mean  :2022-05-18    Mean   : 595.0
##                      3rd Qu.:2024-04-11    3rd Qu.: 800.0
##                      Max.   :2026-03-02    Max.   :9700.0
```

```
llista_cartes <- unique(dades_netes$ID_Card)
llista_resultats <- list()

for (carta_id in llista_cartes) {
  df_temp <- dades_netes %>%
    filter(ID_Card == carta_id) %>%
    mutate(Mes_Any = floor_date(Date, "month")) %>%
    group_by(Mes_Any) %>%
    summarise(Preu_Mensual = mean(Price, na.rm = TRUE), .groups = 'drop') %>%
    complete(Mes_Any = seq(min(Mes_Any), max(Mes_Any), by = "month")) %>%
    mutate(Preu_Mensual = na.locf(Preu_Mensual, na.rm = FALSE)) %>%
    mutate(Preu_Mensual = na.locf(Preu_Mensual, fromLast = TRUE)) %>%
    mutate(ID_Card = carta_id)
```

```

    llista_resultats[[as.character(cartas_id)]] <- df_temp
  }

base_dades_mensual <- bind_rows(llista_resultats)

bd_mensual <- base_dades_mensual %>%
  select(ID_Card, Mes_Any, Preu_Mensual)

saveRDS(bd_mensual, "C:/Users/joelm/Downloads/TFG/bd_mensual.rds")

bd_mensual <- bd_mensual %>%
  left_join(dades_netes %>% select(ID_Card, Vintage) %>% distinct(), by = "ID_Card")

```

Càlcul de les mètriques financeres principals

```

metriques_cartes <- bd_mensual %>%
  group_by(ID_Card, Vintage) %>%
  arrange(Mes_Any) %>%
  mutate(
    # Rendiment simple respecte al mes anterior
    Rendiment_Simple = (Preu_Mensual - lag(Preu_Mensual)) / lag(Preu_Mensual),
    # Preu Màxim Històric (per al drawdown)
    Max_Preu = cummax(Preu_Mensual),
    Drawdown = (Preu_Mensual - Max_Preu) / Max_Preu
  ) %>%
  summarise(
    Preu_Inicial = first(Preu_Mensual),
    Preu_Final = last(Preu_Mensual),
    Preu_Maxim = max(Preu_Mensual),

    # Rendiment Acumulat (ROI) -> per a transformar després en cagr
    ROI_perc = ((Preu_Final - Preu_Inicial) / Preu_Inicial) * 100,

    # Volatilitat anualitzada
    Volatilitat_Anualitzada = sd(Rendiment_Simple, na.rm = TRUE) * sqrt(12) * 100,

    # Maximum Drawdown
    Max_Drawdown_perc = min(Drawdown, na.rm = TRUE) * 100,

    # CAGR (rendiments anualitzats)
    Anys = as.numeric(difftime(last(Mes_Any), first(Mes_Any), units = "days")) / 365.25,
    .groups = 'drop'
  ) %>%
  mutate(
    CAGR_perc = ((Preu_Final / Preu_Inicial)^(1 / Anys) - 1) * 100
  )

```

Índexs base 100 vintage vs modernes

```
# Càlcul index base 100 equiponderat

index_mercat <- bd_mensual %>%
  group_by(ID_Card) %>%
  arrange(Mes_Any) %>%
  mutate(
    Index_Individual = (Preu_Mensual / first(Preu_Mensual)) * 100
  ) %>%
  ungroup() %>%
  group_by(Vintage, Mes_Any) %>%
  summarise(
    Index_Base100 = mean(Index_Individual, na.rm = TRUE),
    .groups = 'drop'
  )
```

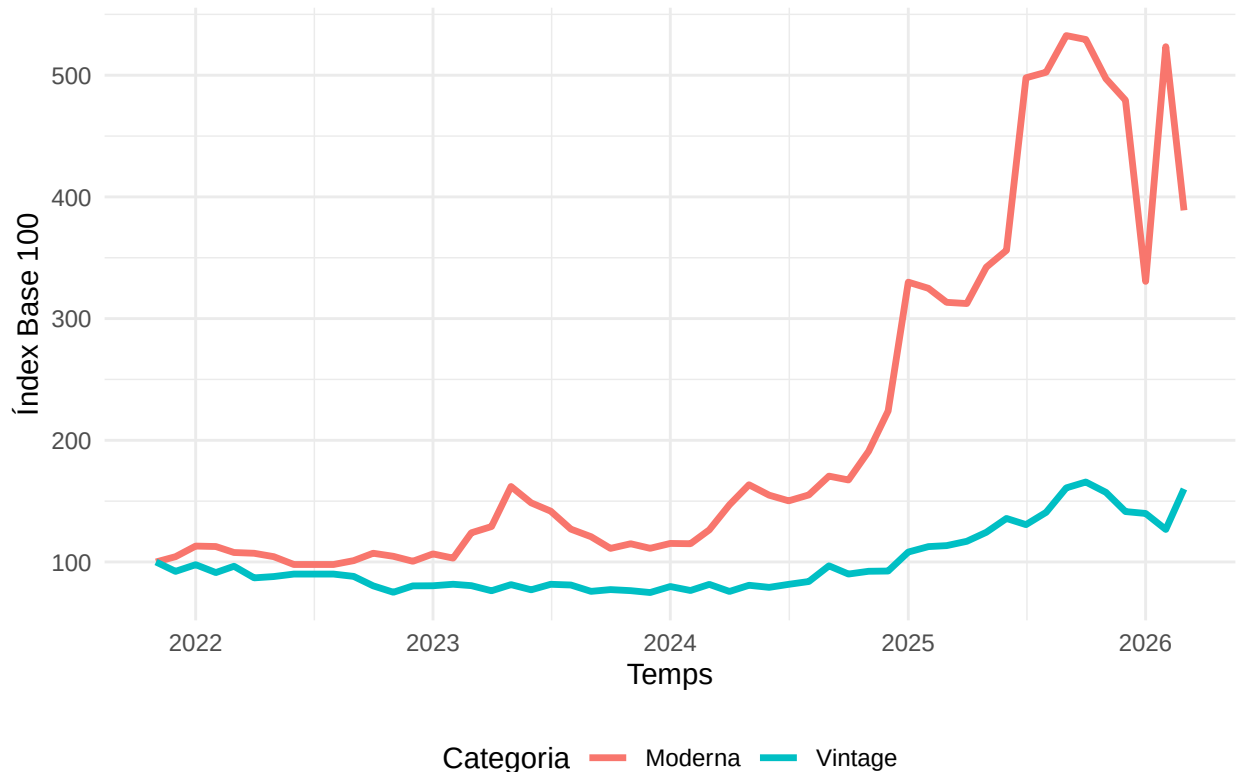
opcio A: tall a 2021 per garantir completitud en la mostra

```
data_inici_estRICTa <- bd_mensual %>%
  group_by(ID_Card) %>%
  summarise(Primera_Data = min(Mes_Any), .groups = 'drop') %>%
  pull(Primera_Data) %>%
  max()

# index base 100
index_mercat_opcioA <- bd_mensual %>%
  filter(Mes_Any >= data_inici_estRICTa) %>%
  group_by(ID_Card) %>%
  arrange(Mes_Any) %>%
  mutate(
    Index_Individual = (Preu_Mensual / first(Preu_Mensual)) * 100
  ) %>%
  ungroup() %>%
  group_by(Vintage, Mes_Any) %>%
  summarise(
    Index_Base100 = mean(Index_Individual, na.rm = TRUE),
    .groups = 'drop'
  )

#grafic
ggplot(index_mercat_opcioA, aes(x = Mes_Any, y = Index_Base100, group = Vintage, color = Vintage)) +
  geom_line(linewidth = 1.2) +
  scale_color_manual(values = c("Vintage" = "#00BFC4", "Moderna" = "#F8766D")) +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  theme_minimal() +
  labs(title = "Evolució de l'Índex Base 100 Equiponderat",
       x = "Temps",
       y = "Índex Base 100",
       color = "Categoria") +
  theme(legend.position = "bottom")
```

Evolució de l'Índex Base 100 Equiponderat



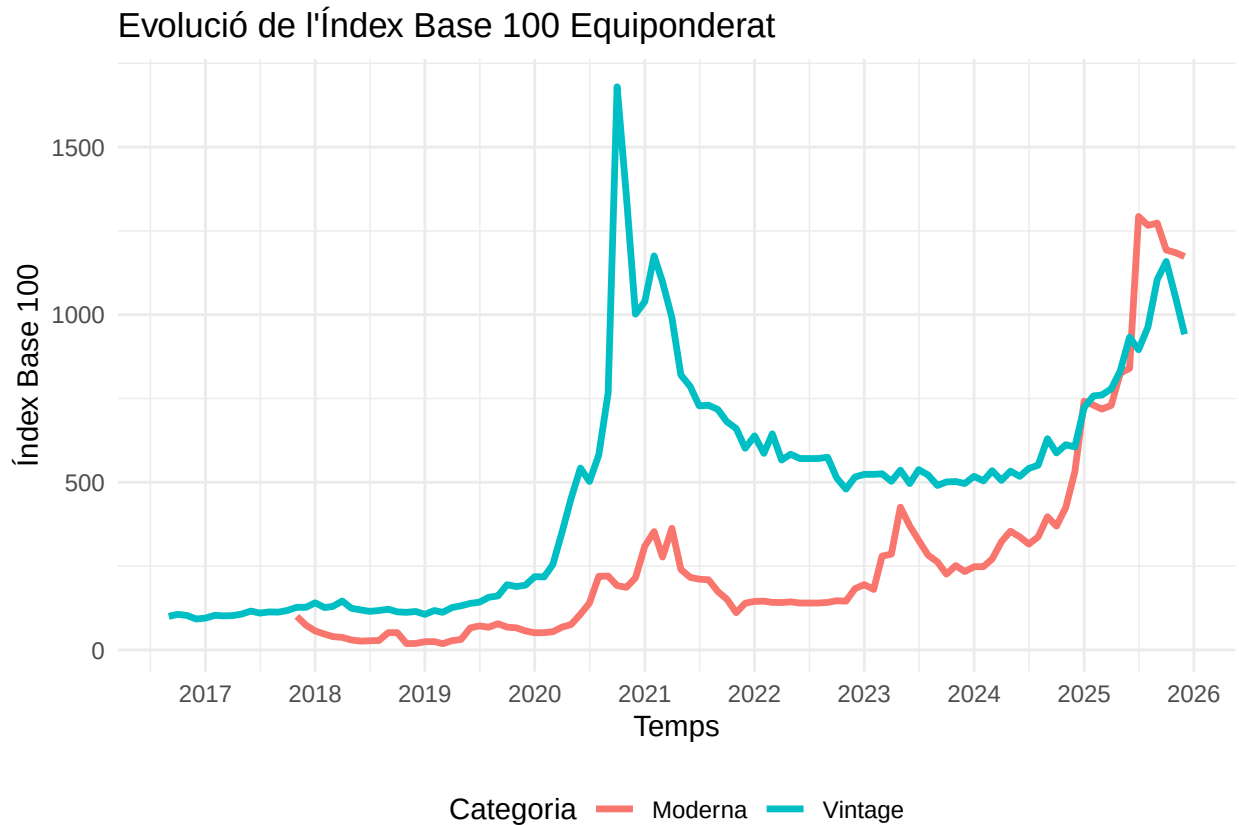
opció B (definitiva): sense talls a les dades, entrades dinàmiques

```
bd_mensual_neta <- bd_mensual %>%
  filter(Mes_Any <= as.Date("2025-12-01"))

# index base 100
index_mercat_opcioB <- bd_mensual_neta %>%
  group_by(ID_Card) %>%
  arrange(Mes_Any) %>%
  mutate(
    Index_Individual = (Preu_Mensual / first(Preu_Mensual)) * 100
  ) %>%
  ungroup() %>%
  group_by(Vintage, Mes_Any) %>%
  summarise(
    Index_Base100 = mean(Index_Individual, na.rm = TRUE),
    .groups = 'drop'
  )

#grafic
ggplot(index_mercat_opcioB, aes(x = Mes_Any, y = Index_Base100, group = Vintage, color = Vintage)) +
  geom_line(linewidth = 1.2) +
  scale_color_manual(values = c("Vintage" = "#00BFC4", "Moderna" = "#F8766D")) +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
```

```
theme_minimal() +
labs(title = "Evolució de l'Índex Base 100 Equiponderat",
     x = "Temps",
     y = "Índex Base 100",
     color = "Categoria") +
theme(legend.position = "bottom")
```



4.2. Correlació amb opcions d'inversió populars

```
# definir actius: S&P 500, Or, Bitcoin
tickers <- c("^GSPC", "GC=F", "BTC-USD")

dades_externes <- tq_get(tickers,
                        get = "stock.prices",
                        from = "2016-01-01",
                        to = "2025-12-31")

# passem les dades a mensuals ja que estan en diàries. agafem directament el tancament mensual
externes_mensual <- dades_externes %>%
  group_by(symbol) %>%
  tq_transmute(select = adjusted,
               mutate_fun = to.monthly,
```

```

        indexAt      = "lastof") %>%
ungroup() %>%
mutate(Mes_Any = as.Date(paste0(format(date, "%Y-%m"), "-01"))) %>%
select(symbol, Mes_Any, adjusted) %>%
group_by(symbol) %>%
arrange(Mes_Any) %>%
mutate(
  Rendiment = (adjusted - lag(adjusted)) / lag(adjusted)
) %>%
ungroup()

retorns_macro <- externes_mensual %>%
  select(Mes_Any, symbol, Rendiment) %>%
  pivot_wider(names_from = symbol, values_from = Rendiment) %>%
  rename(SP500 = `GSPC`, Gold = `GC=F`, Bitcoin = `BTC-USD`)

# creem les columnes amb rendiments de cartes vintage i modernes
retorns_pokemon <- index_mercat_opcioB %>%
  group_by(Vintage) %>%
  arrange(Mes_Any) %>%
  mutate(
    Rendiment = (Index_Base100 - lag(Index_Base100)) / lag(Index_Base100)
  ) %>%
  ungroup() %>%
  select(Mes_Any, Vintage, Rendiment) %>%
  pivot_wider(names_from = Vintage, values_from = Rendiment) %>%
  rename(Poke_Moderna = Moderna, Poke_Vintage = Vintage)

# totes les dades en una
bd_correlacions <- inner_join(retorns_pokemon, retorns_macro, by = "Mes_Any") %>%
  drop_na()

# preparem matriu correlacio
matriu_dades <- bd_correlacions %>% select(-Mes_Any)
matriu_corr <- cor(matriu_dades, method = "pearson", use = "complete.obs")

print(round(matriu_corr, 3))

```

```

##           Poke_Vintage Poke_Moderna SP500 Gold Bitcoin
## Poke_Vintage      1.000      0.087 -0.037 0.031  0.140
## Poke_Moderna      0.087      1.000  0.179 0.163  0.210
## SP500             -0.037      0.179  1.000 0.125  0.374
## Gold              0.031      0.163  0.125 1.000  0.115
## Bitcoin           0.140      0.210  0.374 0.115  1.000

```

Comparació evolucions: Índex Base 100

```

index_macro <- externes_mensual %>%
  filter(Mes_Any >= min(index_mercat_opcioB$Mes_Any)) %>%
  group_by(symbol) %>%
  arrange(Mes_Any) %>%

```

```

mutate(
  Index_Base100 = (adjusted / first(adjusted)) * 100,
  Categoria = case_when(
    symbol == "^GSPC" ~ "S&P 500",
    symbol == "GC=F" ~ "Or",
    symbol == "BTC-USD" ~ "Bitcoin"
  )
) %>%
ungroup() %>%
select(Mes_Any, Categoria, Index_Base100)

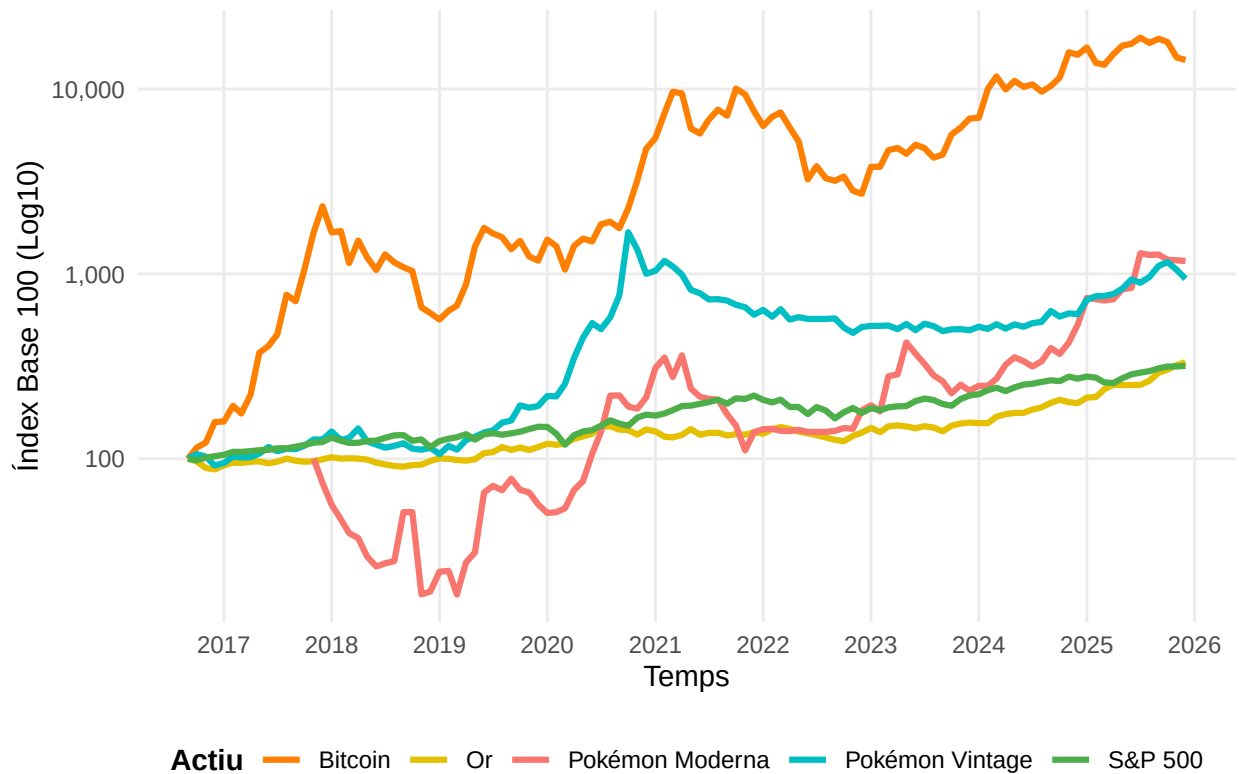
index_pokemon_plot <- index_mercat_opcioB %>%
  rename(Categoria = Vintage) %>%
  mutate(Categoria = paste("Pokémon", Categoria)) %>%
  select(Mes_Any, Categoria, Index_Base100)

dades_grafic_global <- bind_rows(index_pokemon_plot, index_macro)

# generacio grafic
ggplot(dades_grafic_global, aes(x = Mes_Any, y = Index_Base100, color = Categoria)) +
  geom_line(linewidth = 1.1) +
  scale_color_manual(values = c(
    "Pokémon Vintage" = "#00BFC4",
    "Pokémon Moderna" = "#F8766D",
    "S&P 500" = "#4daf4a",
    "Or" = "#e4c000",
    "Bitcoin" = "#ff7f00"
  )) +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  scale_y_log10(labels = comma_format()) +
  theme_minimal() +
  labs(title = "Evolució Global: Col·leccionisme vs. Actius Tradicionals i Digitals",
    x = "Temps",
    y = "Índex Base 100 (Log10)",
    color = "Actiu") +
  theme(
    legend.position = "bottom",
    legend.title = element_text(face = "bold"),
    panel.grid.minor = element_blank()
  )

```

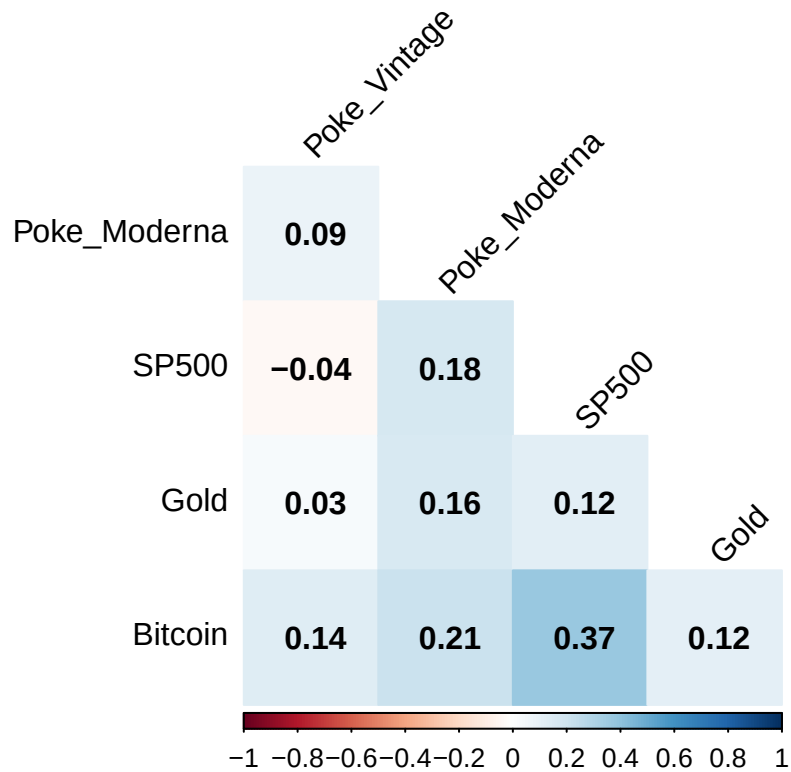
Evolució Global: Col·leccionisme vs. Actius Tradicionals i Digitals



Matriu de correlacions de Pearson

```
# Generem el gràfic visual
corrplot(matriu_corr,
  method = "color",
  type = "lower",
  addCoef.col = "black", # Mostra els nombres dins els quadrats
  tl.col = "black", tl.srt = 45,
  diag = FALSE,
  title = "Matriu de Correlació: Actius de Col·leccionisme vs Tradicionals",
  mar = c(0,0,2,0))
```

Matriu de Correlació: Actius de Col·leccionisme vs Tradicionals

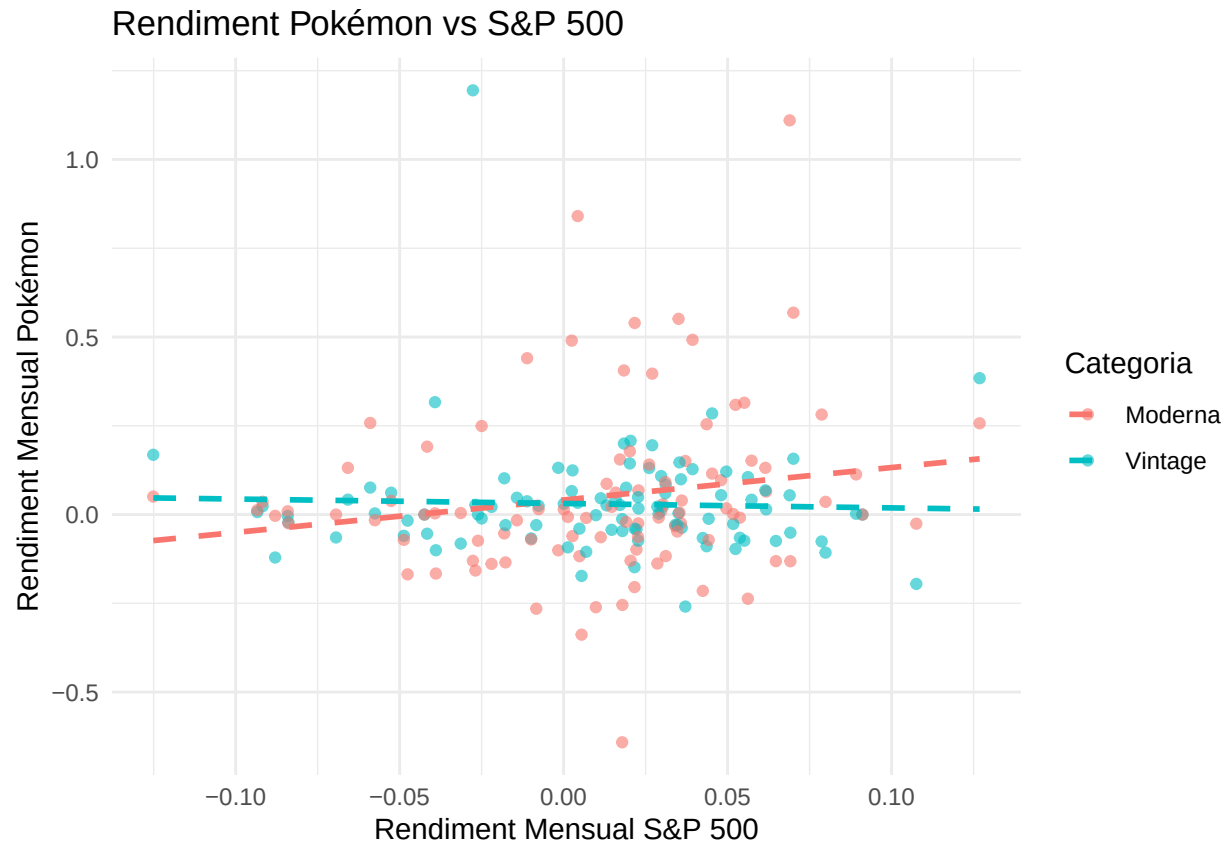


Gràfics de dispersió

vs S&P 500

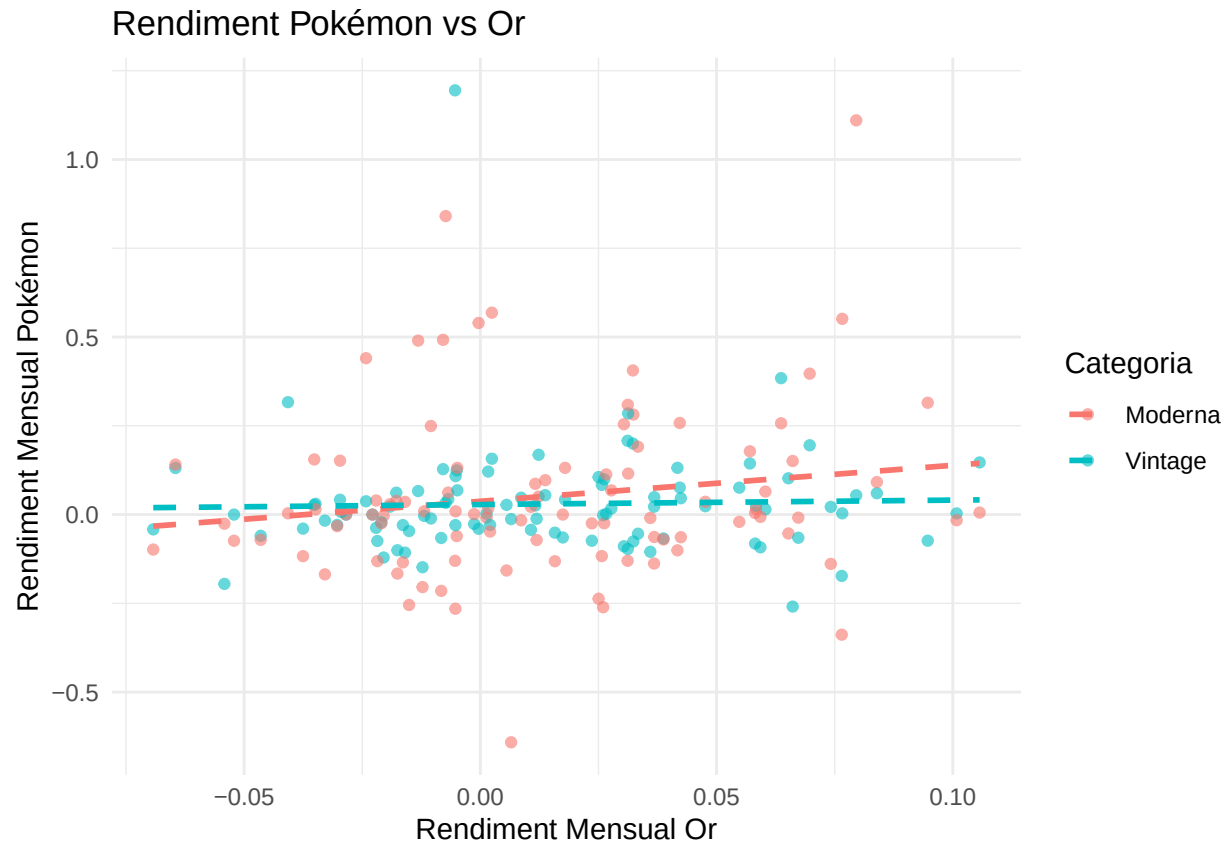
```
dispersion_data <- bd_correlacions %>%
  pivot_longer(cols = c(Poke_Vintage, Poke_Moderna), names_to = "Categoria_Pokemon", values_to = "Rendiment")

ggplot(dispersion_data, aes(x = SP500, y = Rendiment_Pokemon, color = Categoria_Pokemon)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, linetype = "dashed") +
  scale_color_manual(values = c("Poke_Vintage" = "#00BFC4", "Poke_Moderna" = "#F8766D"),
    labels = c("Moderna", "Vintage")) +
  theme_minimal() +
  labs(title = "Rendiment Pokémon vs S&P 500",
    x = "Rendiment Mensual S&P 500",
    y = "Rendiment Mensual Pokémon",
    color = "Categoria")
```



vs Or

```
ggplot(dispersion_data, aes(x = Gold, y = Rendiment_Pokemon, color = Categoria_Pokemon)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, linetype = "dashed") +
  scale_color_manual(values = c("Poke_Vintage" = "#00BFC4", "Poke_Moderna" = "#F8766D"),
    labels = c("Moderna", "Vintage")) +
  theme_minimal() +
  labs(title = "Rendiment Pokémon vs Or",
    x = "Rendiment Mensual Or",
    y = "Rendiment Mensual Pokémon",
    color = "Categoria")
```



vs Bitcoin

```
ggplot(dispersion_data, aes(x = Bitcoin, y = Rendiment_Pokemon, color = Categoria_Pokemon)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, linetype = "dashed") +
  scale_color_manual(values = c("Poke_Vintage" = "#00BFC4", "Poke_Moderna" = "#F8766D"),
    labels = c("Moderna", "Vintage")) +
  theme_minimal() +
  labs(title = "Sensibilitat del Mercat: Rendiment Pokémon vs Bitcoin",
    x = "Rendiment Mensual Bitcoin",
    y = "Rendiment Mensual Pokémon",
    color = "Categoria")
```

Sensibilitat del Mercat: Rendiment Pokémon vs Bitcoin

